

osteosarcoma-mets-dll3-h7r2

Plain-language summary for patient and caregiver

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What this page is

You (or your clinician) asked Libby to look at one specific situation: **metastatic osteosarcoma where the cancer is showing signs that DLL3 — a protein that some new immunotherapies target — might be present**. Your input was that you'd consider high-risk options if the upside is meaningful.

The DLL3 information you gave us was at the **RNA level**— that tells us the gene is being read, but it doesn't tell us whether the DLL3 protein is actually sitting on the surface of the cancer cells (which is what those new drugs need to grab onto). So the recommendations come in two parts: **a first step everyone agreed on** (a confirmatory test), and then a **ranked list of treatment options**— one of which depends on that test coming back positive, and two of which apply regardless.

The first step everyone agreed on

Get a DLL3 protein test on your tumor — specifically an IHC stain using an antibody called SP347. Your oncologist can order this on stored tissue from a previous biopsy, or fresh tissue if needed. It's non-toxic, takes 1–3 weeks, and costs almost nothing compared to a treatment cycle. **All five reviewers ranked this as the first action — without dissent.**

The test result decides whether the lead treatment option (Option 1 below) is on the table. The other two options (Options 2 and 3) apply either way because they don't depend on DLL3 status.

The options the board considered

Option 1 — Tarlatamab via clinical trial NCT06788938

This option is only available if the DLL3 IHC test comes back positive. If the test is negative or below threshold, this option is off the table; Options 2 and 3 below still apply because they don't depend on this test.

The board recommended this as the lead option *when the test is positive*.

What it is. A "bispecific T-cell engager" given as an infusion every two weeks. One end of the molecule grips DLL3 on the cancer cell, the other end recruits an immune T cell to attack it. Currently approved for small-cell lung cancer. The trial accepts patients with **any** DLL3-positive tumor, including osteosarcoma, **as long as the IHC test confirms DLL3 is on the cell surface at sufficient levels.**

What the upside might look like. In small-cell lung cancer, where this drug is approved, tarlatamab nearly halved the risk of death compared to chemotherapy in a recent randomized trial. About 40 of every 100 patients had measurable tumor shrinkage. **Whether this translates to osteosarcoma is exactly the question the trial would answer — and you'd be helping to answer it.**

The main risks. A reaction called "cytokine release syndrome" happens in about half of patients (mostly mild; about 1 in 100 are severe). Neurologic side effects in roughly 10%. The first cycle requires being in hospital for monitoring.

Where the board agreed and where they didn't.

- *Risktaker, advocate, conservative, consensus reviewer*: endorse. The
- *Critic*: still dissents — even with a positive IHC, no osteosarcoma

If the IHC is negative, this option closes entirely. Talk to your oncologist about Option 2 (regorafenib) as the de facto first treatment in that case, and ask about referral to an academic sarcoma center to look for other osteosarcoma trials that don't require DLL3.

Option 2 — Regorafenib

The board recommended this with universal endorsement. This option does not depend on the DLL3 test result.

What it is. A pill at home, 3 weeks on / 1 week off.

What the upside might look like. Two clinical trials in metastatic osteosarcoma showed regorafenib delays cancer growth by about 2 months on average compared to placebo. Neither trial showed a survival improvement (both let placebo patients cross over to active drug, which dilutes survival comparisons). The benefit is real but modest.

The main risks. Skin reaction on hands and feet (the most common limiting side effect), high blood pressure, fatigue, mucositis, mild liver enzyme elevations. About 7 of every 10 patients have at least one moderate-to-severe side effect; almost all are managed by lowering the dose. No treatment-related deaths in either trial.

Why it's ranked below the trial when the trial is reachable. You said you preferred trial-based options. If the trial is reachable (positive IHC

- available slot), the trial gets rank 1; regorafenib is the off-trial

backbone. If the trial is closed (negative IHC, slot unavailable, or intolerable to wait), regorafenib is the de facto first treatment.

Option 3 — Cabozantinib

Same role: backup if regorafenib doesn't work. Universal endorsement; biomarker-independent.

What it is. Another pill at home; targets different growth pathways (MET and VEGFR2) than regorafenib.

What the upside might look like. A 42-patient trial showed about 12 of every 100 patients had tumor shrinkage; about a third had no growth at 6 months. Smaller effect than regorafenib's PFS gain.

The main risks. Similar profile to regorafenib (high blood pressure, hand-foot reaction) plus a notable risk of blood clots in about 7% of patients — relevant for someone with metastatic disease.

When to consider it. If regorafenib is contraindicated, has been used, or wasn't tolerated.

What the board could not figure out

Several things weren't in the inputs:

- **Was DLL3 measured at the protein level (IHC) or only RNA?** Resolved by
- **What treatment have you already had?** The recommendations assume
- **What's your performance status (how active are you day-to-day)?** Most
- **Where do you live, and what's your access to academic cancer centers?**

Questions to ask your oncologist

1. **Can we order DLL3 IHC (SP347 antibody) on my tumor?** That's the first
2. **If the IHC comes back positive, what's the path to NCT06788938?**
3. ****If the IHC comes back negative, what other osteosarcoma trials might**
4. ****If we pursue regorafenib (in either case), what's the plan for**
5. **Have I already had regorafenib or cabozantinib? What was the response?**
6. ****What's the realistic time-to-response for each option, and how will we**
7. ****Is there anything in my labs or scans that would change which of these**
8. ****If the tarlatamab trial is on the table, what's the realistic timeline**

Where to read more

- [Clinician summary](#)— the technical version of this page.
- [Trial table](#)— the trials Libby reviewed.
- [Evidence list](#)— the clinical and lab studies cited.
- [Tumor-board transcript](#)— what each reviewer actually said.
- [Recommendations detail](#)— full structured tables.

Sources

The recommendations cited the following published reports and clinical-trial records:

- Davis et al. *J Clin Oncol*2019 — regorafenib SARC024 RCT in metastatic
- Duffaud et al. *Lancet Oncol*2019 — regorafenib REGOBONE RCT (PMID 30477937)
- Italiano et al. *Lancet Oncol*2020 — cabozantinib CABONE phase-2 in
- Ahn et al. *N Engl J Med*2023 — tarlatamab DeLLphi-301 in SCLC (PMID 37861218)
- Mountzios et al. *N Engl J Med*2025 — tarlatamab DeLLphi-304 in SCLC
- Yao et al. *Oncologist*2022 — DLL3 as a therapeutic target review
- Wang et al. *J Exp Clin Cancer Res*2018 — Notch3/DLL3 axis in osteosarcoma
- Trial registry NCT06788938 — UCCC-01 / UCLA L-10 tarlatamab basket